Relationship Between Color Characteristics and Plant Growth in Pennycress Plants

1. Introduction

My team and I were presented with a dataset that includes information from an experiment at the Oak Ridge laboratories. The data is from a pennycress plant experiment where they grew different genotypes of pennycress inside. These plants were treated with 6 different levels of nickel exposure: 0, 50, 100, 200, 500 and 1,000. These plants were on a conveyor belt that was on a loop to go under a camera where pictures of the plants were taken. The pictures went through a masking program that isolated the pixels of the plant from the pixels of the background. The plant pixels were then evaluated to extract information such as the area of the plant and the mean pixel intensity for red, green and blue color values. After looking through some of the data the pictures provided, we came up with the following questions:

- How do plant color characteristics change over time?
- Is plant color associated with plant growth?
- Can plant color features explain variation in plant growth?
- Can color and shape features identify stressed or unhealthy plants?
- What is the optimal 3-to-5-day harvest window?
- Can we determine whether a plant will thrive after a short growth period based on color?

2. Dataset

We were given two datasets, rgb_design and rgb_features. The information we used from rgb_design includes a sample_key, plant_id, date, genotype, and treatment. The rgb_features dataset included sample_key, plant_id, date, area_px, red_mean, green_mean, blue_mean and ngrdi_mean. The sample_key column included a specific key that each unique observation was given. In this dataset, each observation was a picture taken of the plant as it went through the conveyor belt on a given day. Thus, we were able to merge the datasets based on the sample_key to compile the image data.

3. Research

To ensure we fully understood the information, we had to research what the columns for each of the colors meant. We found out that the red_mean, green_mean and blue_mean are each the mean pixel intensity for a specific color in one image. Also, we learned that the ngrdi_mean was the mean Normalized Red Green Difference Index for one image. This index is calculated by:

$$NGRDI = \frac{Green - Red}{Green + Red}$$

The NGRDI is used in a variety of agricultural applications, but for our purposes, it is used for monitoring plant health. It uses the fact that plants absorb visible light and reflect near-infrared radiation, and contrasts them. The index is a value between -1 and 1, when it is close to -1. In our application, a value close to -1 indicates necrosis. There is little to no chlorophyll left in the plant, and it will appear red or brown. A value close to 0 will indicate chlorosis or bleaching. The plant will appear yellow or brown, and are early signs of a dying plant. A value close to 1 indicates a healthy, green plant.

We also found an article explaining an experiment where they used image masking and color values to examine the lettuce plants they were growing (Kang et al., 2025). After reading through this article, we found 3 new indices that may be helpful for our purposes. Those indices are excess green (EXG), excess red (EXR) and visible atmospherically resistant index (VARI). They are calculated as follows:

$$r = \frac{Red}{Red + Blue + Green} \quad b = \frac{Blue}{Red + Blue + Green} \quad g = \frac{Green}{Red + Blue + Green}$$

$$EXG = 2 * g - b - r$$

$$EXR = 1.4r - g$$

$$VARI = \frac{g - r}{g + r - b}$$

The EXG index is used to determine chlorophyll concentration, and has a theoretical range of -1 to 2. However, in our case, the value is typically around 0.1 to 0.3 for a healthy plant and below 0 for a dead or dying plant. The EXR index is used to detect vegetative stress and has a theoretical range of -1 to 1.4. In our case, it is typically close to, but below 0 for a healthy plant and above 0 for a stressed

plant. The VARI index is very similar to the NGRDI index mentioned above. However, the VARI accounts for glare or self-shadowing with its blue term.

4. Exploratory Data Analysis

To start off, we looked at how the rgb values changed over time. To accomplish this, we took the mean rgb pixel intensity for each day and graphed each color as a separate line over time. We also took the mean area for each day and graphed that over time, and stacked the plots for easy comparison.

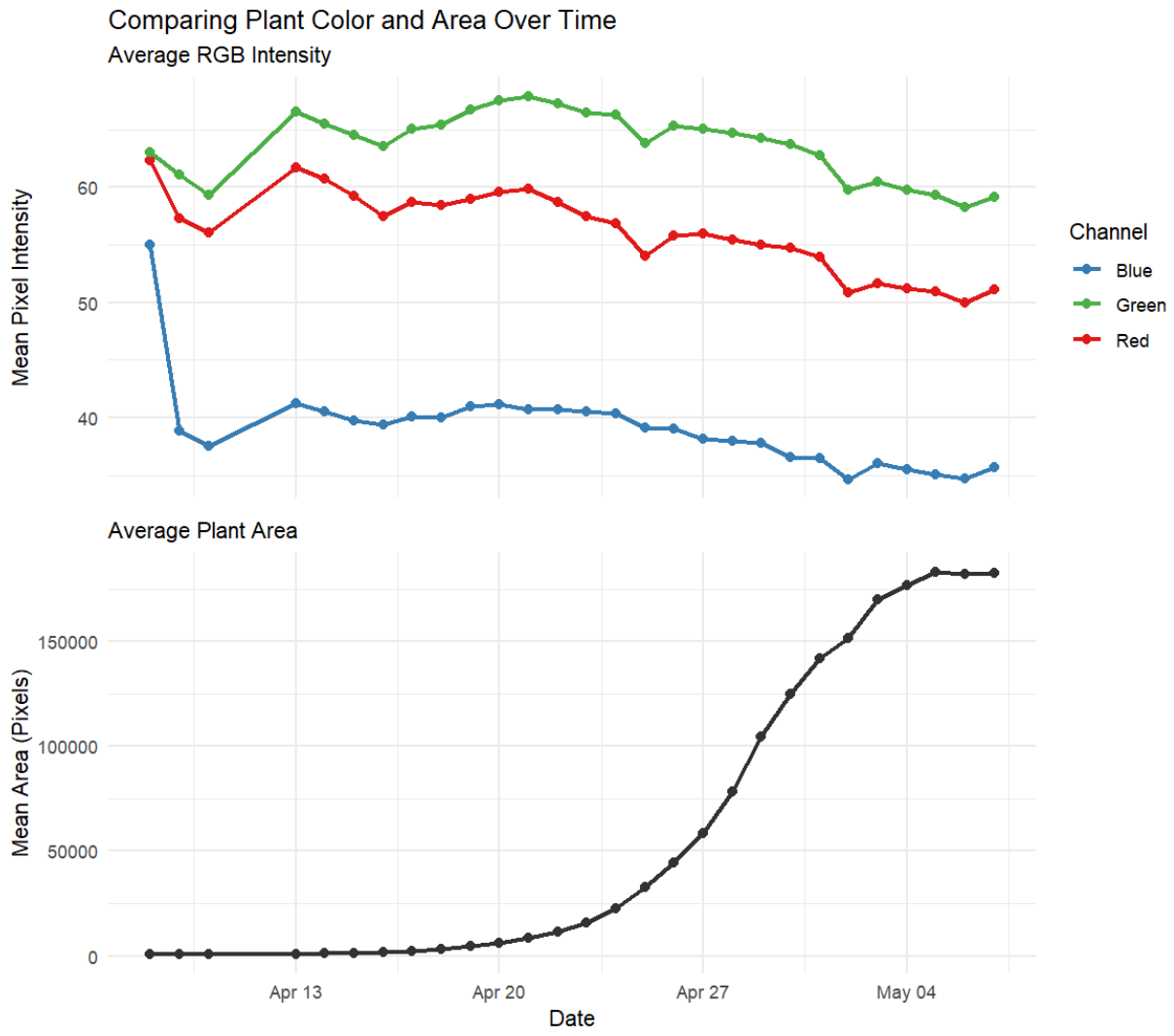


Figure 1: Plant Color and Area Over Time

From this graph, we found something unexpected. We originally believed that as the plant grew, and the area increased, the colors would increase. We at least thought that the green line would increase. However, the opposite occurred. As the

area increases, the colors actually decrease. We figured out that this is due to self-shadowing. As the plant grows, the leaves “stack” on top of each other, and make the leaves below them look darker. This decreases the pixel color intensity. Which lead us to want to know if a specific color is more intense if a plant is dead or dying. This meant we needed to figure out which plants were considered dead. We then formed a graph of the maximum area each plant reached, and ranked them from smallest to largest.

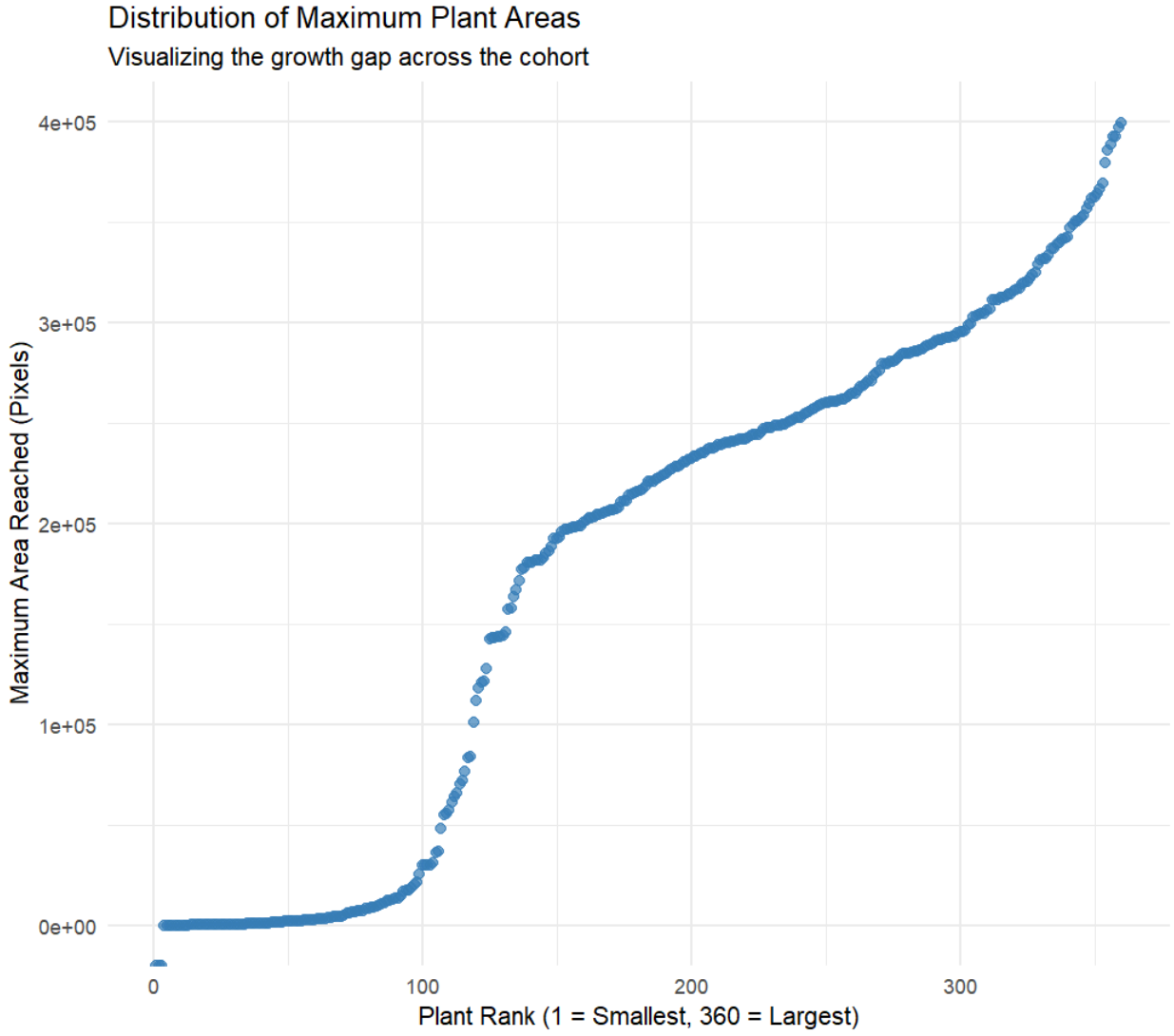


Figure 2: Maximum Plant Area Ranked

There is a gap around rank 100 where the line thins out and there are not as many plants reaching those areas. Thus, we decided to count any plant below a threshold of 20,000 pixels (the area at about rank 100) as a dead plant. This resulted in classifying 264 alive plants and 96 dead plants. We then took those two

groups and compared their mean daily color values over time. These graphs were overlaid to allow for easier comparison.

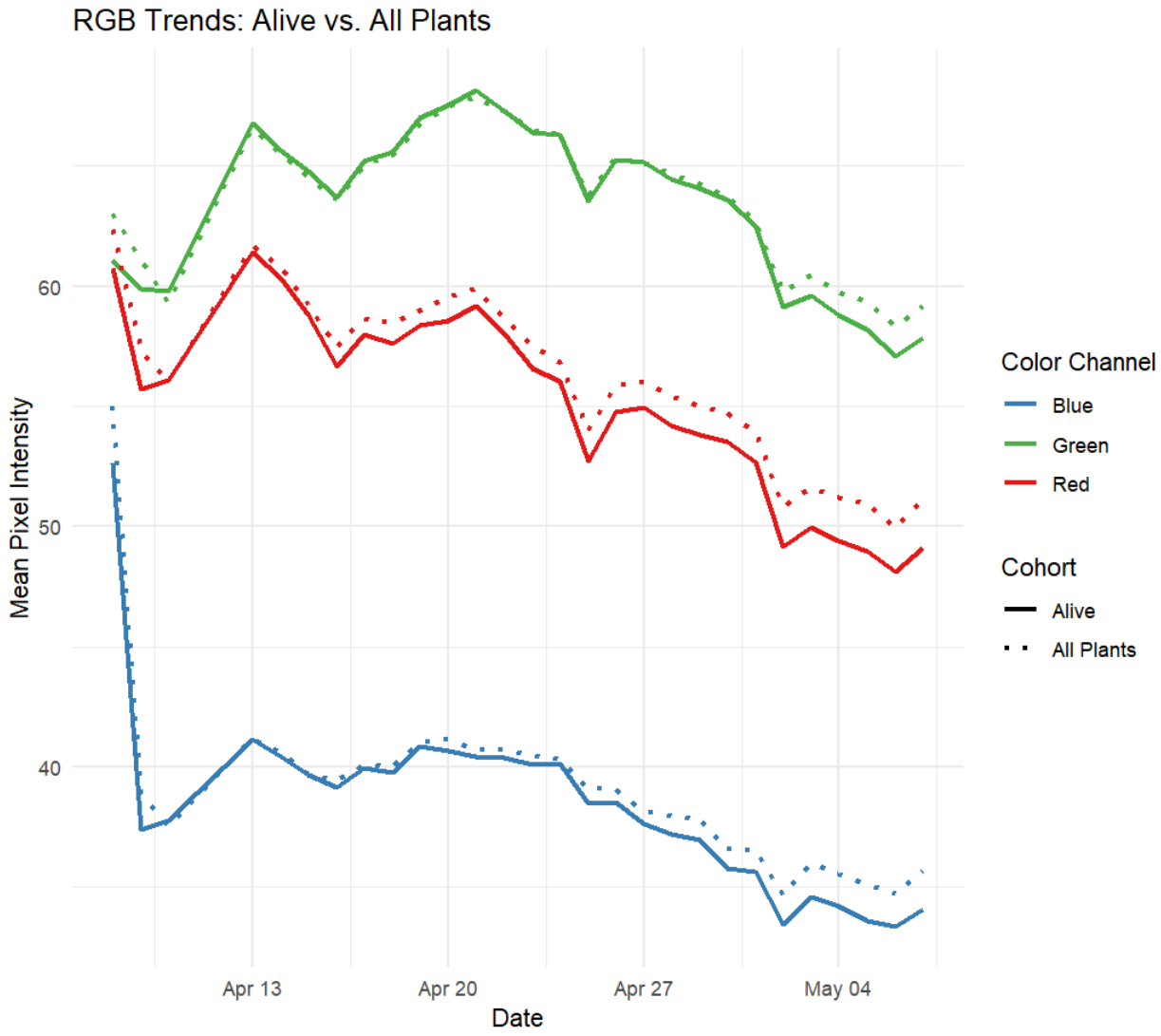


Figure 3: Color Values Over Time for Alive versus All plants

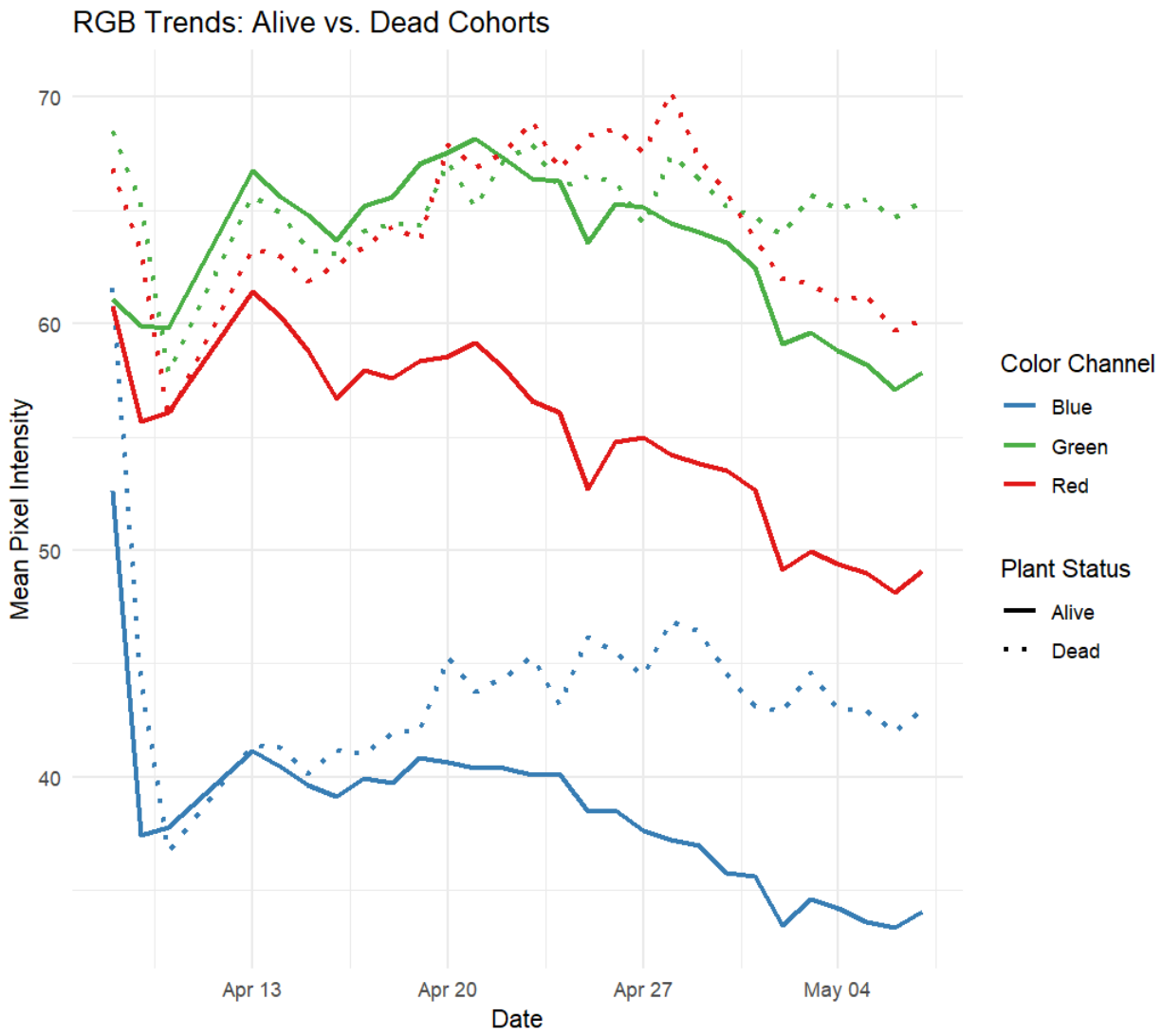


Figure 4: Color Values Over Time for Alive versus Dead plants

The red line for all plants in Figure 3 is strictly above the line for the alive plants after day 5 (April 13th). However, the red line for the dead plants was drastically higher than either of the other red lines. The same conclusion can be reached with the blue line. The green line is different. The dead plant line oscillates around the alive plant line, and same for the all plants line, although much less noticeable. This may indicate that we can predict whether a plant will live or die by day 5. This will have to be explored later on.

```
[1] "--- Summary Statistics: WHOLE DATASET ---"
> df_classified %>%
+   select(red_mean, green_mean, blue_mean) %>%
+   summary() %>%
+   print()
    red_mean        green_mean       blue_mean
 Min.   : 25.30   Min.   : 26.50   Min.   : 11.29
 1st Qu.: 51.01   1st Qu.: 59.78   1st Qu.: 34.65
 Median : 55.65   Median : 64.47   Median : 38.33
 Mean   : 56.20   Mean   : 64.18   Mean   : 38.68
 3rd Qu.: 59.75   3rd Qu.: 68.49   3rd Qu.: 42.21
 Max.   :130.37   Max.   :114.36   Max.   :104.48
 NA's   :4690     NA's   :4690     NA's   :4690
```

```
[1] "--- Summary Statistics: ALIVE PLANTS ---"
> df_classified %>%
+   filter(status == "Alive") %>%
+   select(red_mean, green_mean, blue_mean) %>%
+   summary() %>%
+   print()
    red_mean        green_mean       blue_mean
 Min.   :28.66   Min.   :31.16   Min.   :17.68
 1st Qu.:50.69   1st Qu.:59.89   1st Qu.:34.40
 Median :55.17   Median :64.37   Median :37.93
 Mean   :55.16   Mean   :64.07   Mean   :38.09
 3rd Qu.:58.94   3rd Qu.:68.27   3rd Qu.:41.72
 Max.   :98.10   Max.   :98.75   Max.   :83.80
 NA's   :629     NA's   :629     NA's   :629
```

```
[1] "--- Summary Statistics: DEAD PLANTS ---"
> df_classified %>%
+   filter(status == "Dead") %>%
+   select(red_mean, green_mean, blue_mean) %>%
+   summary() %>%
+   print()
    red_mean        green_mean       blue_mean
 Min.   : 25.30   Min.   : 26.50   Min.   : 11.29
 1st Qu.: 55.21   1st Qu.: 58.71   1st Qu.: 37.65
 Median : 61.90   Median : 65.36   Median : 41.62
 Mean   : 63.97   Mean   : 65.05   Mean   : 43.08
 3rd Qu.: 70.55   3rd Qu.: 71.02   3rd Qu.: 46.62
 Max.   :130.37   Max.   :114.36   Max.   :104.48
 NA's   :4061     NA's   :4061     NA's   :4061
```

Figure 5: Summary Statistics for Whole Dataset (Top Left), Alive Plants (Top Right), and Dead Plants (Bottom Left)

We took the datasets that were split by plant status and ran some summary statistics on them. These summary statistics are based on individual observations, not averages. So, the max red value of 130.37 is the highest value a single plant reached. First, we noticed that the difference between the min and the max of rgb values was exactly the same for the dead plants as well as the whole dataset. Additionally, we noticed that the range is much smaller for the alive plants dataset. The differences between the rgb values are also much closer together between each color for the alive plants with a difference between red, green and blue only being about 4. Whereas the difference between red, green and blue for the dead plants is about 20. Through a little bit of research, we learned that the likely reason the living plants have such a tighter range is due to the chlorophyll regulation. When a plant is alive, it can accurately regulate its chlorophyll to allow for the proper amount of sunlight to be captured. However, when a plant is dying or dead, it can no longer regulate its colors, allowing for a larger range. We thought it was odd

that the dead plants have both the largest and the smallest numbers, so we had looked into that as well. The high values are likely due to bleaching or drying out when the plant is dying. The low values are likely due to the plant not growing much, so there are not many pixels to average out, or necrosis taking over.

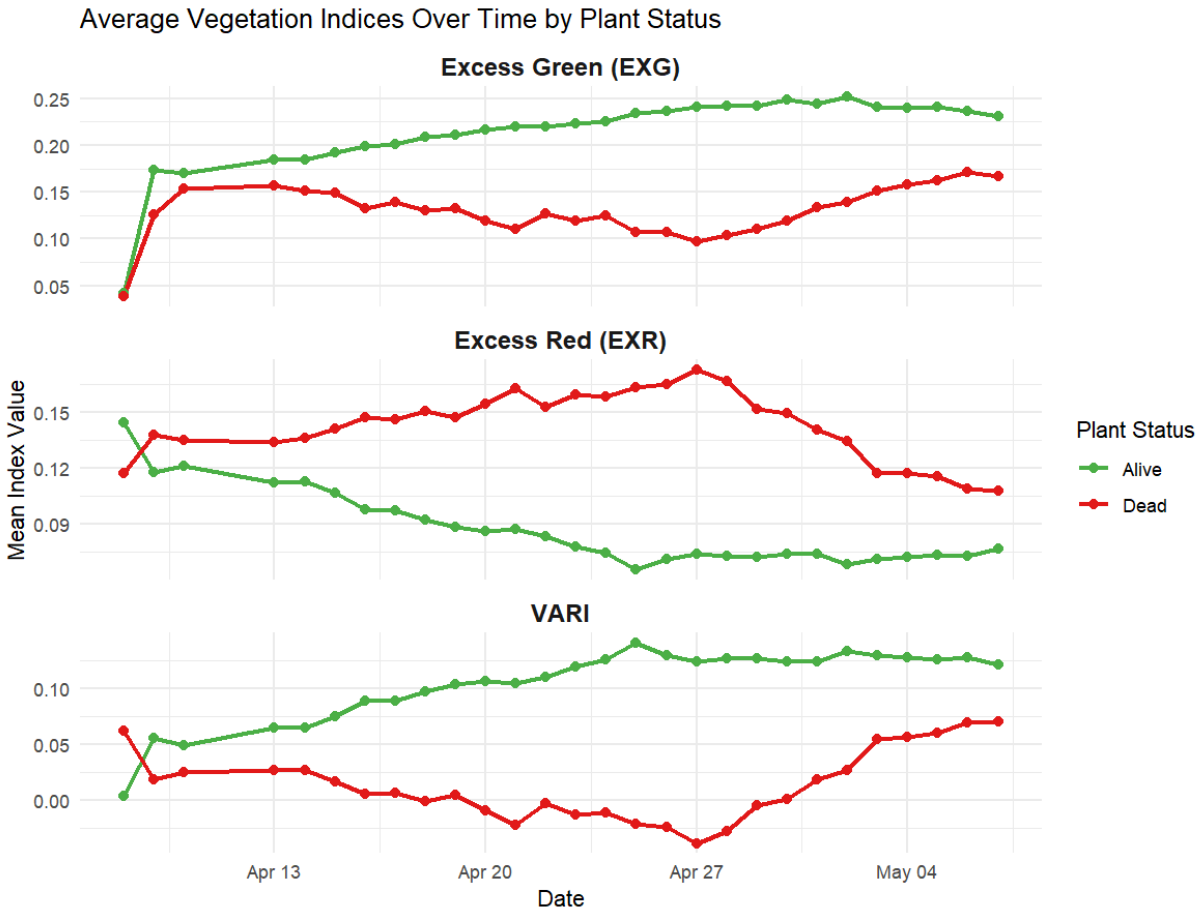


Figure 6: Mean Daily Index Values Over Time by Plant Status

We calculated the EXG, EXR and VARI indices mentioned before and graphed their daily averages each separately over time. We also created two separate lines on the graph for the living plants versus the dead ones to compare their behavior. The EXG is significantly higher for the living plants than the dead ones. The difference starts getting even wider by April 13. The dead plants do increase in concentration towards the end of the month, but never pass the alive plants. The EXR is significantly lower for the living plants than the dead ones. This makes sense since the EXR shows vegetative stress. It is interesting that they start out with the living plants higher than the dead ones, but after a day or two, it switches

and they stay with the dead ones higher. By April 14-15, the values start separating even further. The VARI is significantly higher for the living plants than the dead ones. The difference gets increasingly wider starting around Apr 14-15. Just like with the EXR, the values start with the dead plants being higher than the living ones, but switches after a day or two and stays with the living plants higher. All of the lines start close together, get significantly further apart, and then get close again towards the final week. The lines never cross past day 2 or 3.

```
Call:
glm(formula = is_alive ~ early_EXG + early_EXR + early_VARI +
    early_NGRDI, family = "binomial", data = early_color_predictors)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)    7.519      9.913   0.759    0.448
early_EXG      7.354      7.388   0.995    0.319
early_EXR    -55.121     71.322  -0.773    0.440
early_VARI   -46.036     26.295  -1.751    0.080 .
early_NGRDI   46.452     66.124   0.703    0.482
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 358.75  on 325  degrees of freedom
Residual deviance: 314.39  on 321  degrees of freedom
AIC: 324.39

Number of Fisher Scoring iterations: 5
```

Figure 7: Logistic Regression Model with 4 Predictors

We ran a logistic model using EXG, EXR, VARI and NGRDI as predictors to formally test if any of these indices will help predict if a plant will survive. It was found that none of the indices showed statistical significance. This appeared odd considering each graph prior to this showed a relationship. We believed there was a multicollinearity problem, and ran each of the predictors individually. Through this, we found that each index was highly significant. The collinearity is reasonable since each index is calculated with the same average pixel intensities. EXR was the only index that was negatively related. Which was expected due to a decrease in

EXR values as time went on for the healthy plants. The other 3 indices were positively related, which is corroborated in the graphs previously.

```
[1] "--- Linear Mixed Model: EXG (Greenness) Trajectory ---"
> summary(exg_lmm)
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest'
Formula: EXG ~ day_numeric * status + (1 | plant_id)
   Data: df_color_lmm

REML criterion at convergence: -78778.1

Scaled residuals:
    Min      1Q  Median      3Q     Max
-7.5798 -0.4821  0.0113  0.5388  7.7394

Random effects:
 Groups   Name        Variance  Std.Dev.
 plant_id (Intercept) 0.0007412 0.02722
 Residual             0.0008412 0.02900
Number of obs: 18890, groups:  plant_id, 357

Fixed effects:
                          Estimate Std. Error         df t value Pr(>|t|)
(Intercept)             1.791e-01  1.775e-03  3.859e+02  100.95   <2e-16 ***
day_numeric             2.685e-03  3.271e-05  1.852e+04   82.09   <2e-16 ***
statusDead             -5.105e-02  3.742e-03  5.005e+02  -13.64   <2e-16 ***
day_numeric:statusDead -2.539e-03  9.343e-05  1.888e+04  -27.18   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) dy_nmr sttsDd
day_numeric -0.303
statusDead  -0.474  0.144
dy_nmrc:stD  0.106 -0.350 -0.391
> anova(exg_lmm)
Type III Analysis of Variance Table with Satterthwaite's method
                   Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
day_numeric        0.77221 0.77221     1 18881.5  917.93 < 2.2e-16 ***
status             0.15661 0.15661     1   500.5  186.16 < 2.2e-16 ***
day_numeric:status 0.62150 0.62150     1 18881.5  738.78 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 8: Linear Mixed Model for EXG

Next, we ran a linear mixed model on EXG with “Alive” status as the baseline. This model is used for non-independent measurements by adding fixed and random effects. Fixed effects show the average relationship or trend across the population and random effects account for variation in the groups or individual differences. We found that the EXG index was highly dependent on whether they lived or died. Each group had statistically different values for the EXG index. Under the fixed effects estimate, it was 0.00268, indicating the living plants increased their EXG index by about 0.00268 each day. Additionally, the statusDead

estimate shows that these plants started out 0.051 units lower in EXG index than the living plants did. The estimated change per day for the dead plants was -0.00253 units different from the living plants. Which means that the daily increase in EXG value for the dead plant was 0.00015 per day. Indicating a close to 0 increase in green intensity due to the lack of chlorophyll in dying plants. We wanted to know if any of the other indices would produce more interesting results. The one that told us more than the others was the NGRDI. So we ran a linear mixed model on that one next.

```
[1] "--- Linear Mixed Model: NGRDI Trajectory ---"
> summary(NGRDI_lmm)
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLme
rTest']
Formula: ngrdi_mean ~ day_numeric * status + (1 | plant_id)
   Data: df_color_lmm

REML criterion at convergence: -90993.1

Scaled residuals:
    Min      1Q  Median      3Q     Max
-8.9414 -0.4644  0.0526  0.5266  7.8604

Random effects:
 Groups   Name        Variance  Std.Dev.
 plant_id (Intercept) 0.0004375 0.02092
 Residual             0.0004396 0.02097
Number of obs: 18890, groups:  plant_id, 357

Fixed effects:
                          Estimate Std. Error         df t value Pr(>|t|)
(Intercept)              4.198e-02  1.355e-03  3.825e+02   30.99   <2e-16 ***
day_numeric              2.123e-03  2.364e-05  1.852e+04   89.81   <2e-16 ***
statusDead              -3.932e-02  2.839e-03  4.858e+02  -13.85   <2e-16 ***
day_numeric:statusDead -1.545e-03  6.762e-05  1.889e+04  -22.85   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) dy_nmr sttsDd
day_numeric -0.287
statusDead  -0.477  0.137
dy_nmrc:stD  0.100 -0.350 -0.373
> anova(NGRDI_lmm)
Type III Analysis of Variance Table with Satterthwaite's method
                   Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
day_numeric        0.70207 0.70207     1 18885.9 1596.99 < 2.2e-16 ***
status             0.08437 0.08437     1   485.8  191.91 < 2.2e-16 ***
day_numeric:status 0.22946 0.22946     1 18885.9  521.96 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 9: Linear Mixed Model on NGRDI

The intercept estimate shows that the living plants started at a NGRDI of 0.04198, indicating they started out green. The dead plants started 0.03932 units lower than the living plant did, meaning the dead plants started at a NGRDI of 0.00266. Which is significantly close to 0, indicating that the plants that ended up dying started out stressed, causing them to be equally green and red. This can mean that NGRDI can help us predict whether or not a plant will die early on. The high f-value for day_numeric shows that NGRDI is a good predictor for tracking healthy growth over time. The day_numeric:statusDead in the fixed effects tells us that the dead plants grew 0.001545 slower than the living plants, which results in a 0.000578 growth rate for the dead plants rather than the 0.002123 growth rate of the living plants.

5. Resources

Kang, Z., Zhou, B., Fei, S., & Wang, N. (2025). Predicting the greenhouse crop morphological parameters based on RGB-D computer vision. *Smart Agricultural Technology*, *11*, Article 100968. https://doi.org/10.1016/j.atech.2025.100968